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Chapter 1

Activity in Brownian Motion

1.1 Introduction

The final part of this manuscript is dedicated to the study of active matter and its interplay with Brownian motion. This subject was actually the motivation for my PhD thesis, and the first part of this Chapter presents my first introduction to the field of active matter. I begin with a full introduction to what active matter means, and how it is typically modeled.

The second part, which came much later in my PhD, treats the problem of active Taylor-Aris dispersion. The notion of Taylor-Aris dispersion is described in the previous Chapter. For the sake of clarity, I thus invite the reader to consult its developments before reading this part. We conducted this work in tandem with Marc Lagoin, Guirec de Tournemire, Ahmad Badr and Antoine Allard, and published it in PNAS.

1.1.1 Activity

Authors define active matter as one or both of the following:

1. "energy-consuming particles" : individual units that extract free energy from internal or external sources and convert it to motion, forces, or mechanical stresses (this is the definition I use in this manuscript). This work is not necessarily expressed in the form of self-propelling [1, 2, 3, 4]. They are thus inherently out-of-equilibrium systems;
2. "interacting particles" : individual self-propelled units which interact locally and exhibit emergent, collective behaviour, such as hydrodynamic interactions (swimming microorganisms, squirmers, active droplets...)[3], phoretic interactions (cat-

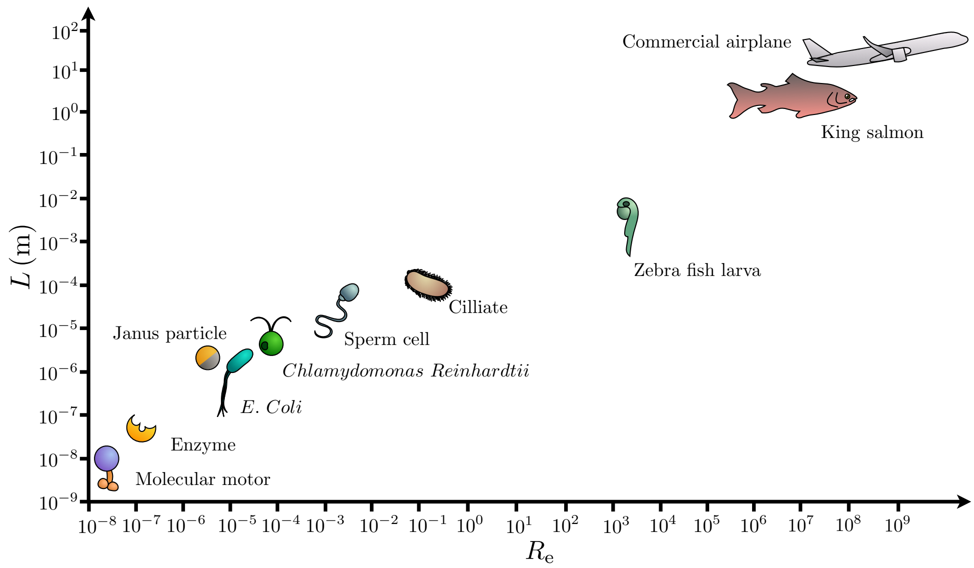

Figure 1.1: Comparison of typical length scales and Reynolds numbers for various active systems.

alytic Janus colloids, some chemotactic bacteria...)[5], mechanical force transmission (cellular tissue, cytoskeleton, active gels...)[1], and even social interactions (human crowds, bird flocks, fish schools, autonomous drones...)[6], to only cite a few. Some of these interactions can be accurately represented with minimal models, which approximate them as alignment interactions with some noise, such as the Vicsek model [7]. Certain authors distinguish these instances of interacting active matter from the first I mentioned [4].

Numerous examples of active systems exist, and these systems can be found at any scale (Fig. 1.1). I find it convenient to not only classify them by size but also by Reynolds number, as most of them will typically use their surrounding energy source to move in space. Interestingly, the definition of active matter is loose enough to even include human beings, robots and even large aircrafts. This thesis articulates around notions of small systems, and I thus refer to these systems form there on.

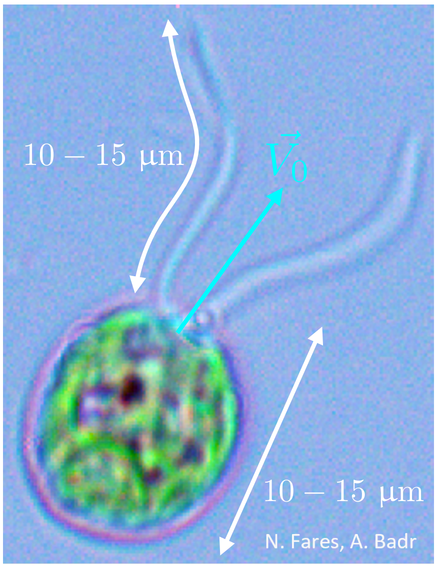

Figure 1.2: Photograph of a *Chlamydomonas reinhardtii* alga taken by Ahmad Badr and Nicolas Fares in our experimental rooms. The body and the flagella each measure about 10µm. The alga also features an eyespot which allows it to sense incoming light and adapt.

1.1.2 *Chlamydomonas reinhardtii* as a model swimmer

A particularly convenient experimental model of a biological microswimmer is the unicellular green alga *Chlamydomonas reinhardtii* (Fig. 1.2) [8]. These tiny algae are the system we use in the study at the end of this Chapter. They measure only a few tens of microns at most. However, through the beating of their anterior flagella, they self-propel in a fluid at surprisingly high speeds of ten body lengths per second. Yet, this is not sufficient for *Chlamydomonas* to be considered high-Reynolds systems ($R_e \gg 1$), as they typically reach a Reynolds number of about $R_e = 1 \times 10^{-4}$. They thus still evolve in a very viscous medium.

Furthermore, *Chlamydomonas reinhardtii* can orient towards or away from light sources by phototaxis, a response mediated by and "eye spot" rhodopsin photoreceptor and controlled by both light intensity and directional sensing [9, 10, 11]. In the absence of a directional light signal, its swimming can display run-and-tumble-like dynamics, with random switches between straight swimming and abrupt reorientation events [8]. Under appropriate illumination and near a solid boundary, cells instead attach through their flagella and perform surface gliding, driven by flagellar surface transport instead ordinary swimming [12, 13, 14, 15, 16]. Because of this simple geometry and well established and studied behaviour, *Chlamydomonas* has become a standard model system for active matter in fluids.

1.2 Modelling approaches

Multiple models exist to study active matter. I describe here the ones I have come to know or use during my thesis, either for published works or personal interest. I also provide in appendix and on the GitHub repository ADD GITHUB a simple Jupyter file to simulate Active Brownian Particles (ABP), Run-And-Tumble particles (RTP) and to compare them to the passive case. This file also contains a very small exploration of chiral ABP.

1.2.1 Active Brownian Particles

For comparison purposes, let us recall that a purely passive colloid undergoes thermal diffusion. This thermal diffusion affects both the translational degrees of freedom (x and y for a two-dimensional case) and the rotational degree of freedoms (ϕ for a two-dimensional case). We define from the Stokes-Einstein relation the translational thermal diffusion coefficient D as:

$$D = \frac{k_{\mathrm{B}}\Theta}{6\pi\eta a} \ , \tag{1.1}$$

where $k_{\mathrm{B}}\Theta$ is the thermal energy of the bath, η the fluid visicosity and a the colloid radius. The rotational diffusion coefficient D_{R}, however, is proportional to the colloid's volume:

$$D_{\mathrm{R}} = \frac{k_{\mathrm{B}}\Theta}{8\pi\eta a^3} = \tau_{\mathrm{R}}^{-1} \ , \tag{1.2}$$

with τ_{R} a characteristic reorientation timescale.

We have already stated that we can describe the motion of the passive colloid with the following Langevin equations for the overdamped limit:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= \sqrt{2D}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= \sqrt{2D}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{R}}}\xi_\phi(t) \ . \end{aligned} \tag{1.3}$$

where $\xi_i(t)$ white Gaussian noises of correlation $\delta(t)$. This process is heavily reliant on the thermal diffusion coefficients D and D_{R}, which are themselves proportional to temperature and inversely proportional to colloid radius and fluid viscosity. This means that thermal Brownian motion is strongly hindered in the case of larger colloids, where “larger” means of the order of a few tens of microns. For example, if we take for comparison with later cases a colloid of a size that is comparable to a *Chlamydomonas*, that is to say $a = 10\mu\text{m}$, we get $D = 2.2 \times 10^{-3} \mu\text{m}^2.\text{s}^{-1}$. We interpret that by pure thermal diffusion, a colloid of this size would need tens of hours to explore an area equivalent to its own size.

Lucky, as we discussed already, swimmers can circumvent this limitation by collecting energy from their surroundings to self propel with a given velocity $\vec{v}$. A very simple model used to represent this scenario is the ABP model. We describe this model with the following Langevin equations:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_\mathrm{R}}\xi_\phi(t) \ . \end{aligned} \tag{1.4}$$

These equations can then be discretised with a very elementary Euler scheme, and I provide on the thesis’ GitHub page a simple code to generate trajectories for this case. The MSD in that case follows [17, 18, 19] (see Fig. 1.5)

$$\boxed{\mathrm{MSD}_x(t) = \mathrm{MSD}_y(t) = 2Dt + \frac{v^2}{D_\mathrm{R}^2}\left(D_\mathrm{R}t + e^{-D_\mathrm{R}t} - 1\right)} \ . \tag{1.5}$$

We thus expect a thermally diffusing regime at very short timelags with diffusivity D, followed by a superdiffusive regime in intermediate times, and finally, after the characteristic timescale τ_R, a second, higher diffusive regime at very long times with a new effective diffusivity D_0, as $D_0 = \frac{v^2}{2D_\mathrm{R}}$ (see Figure 1.3 for a comparison of a passive Brownian colloid and an ABP).

perhaps change D0 to Deff in all. Check later for homogeneisation.

Chiral ABPs

Some microswimmers or clusters of microswimmers have irregular shapes or asymmetrical propulsion mechanisms. It is possible to model the consequences of these assymetries by adding a chiral component to the ABP model.

While the ABP translation mechanisms remain unchanged, we add a constant rotation

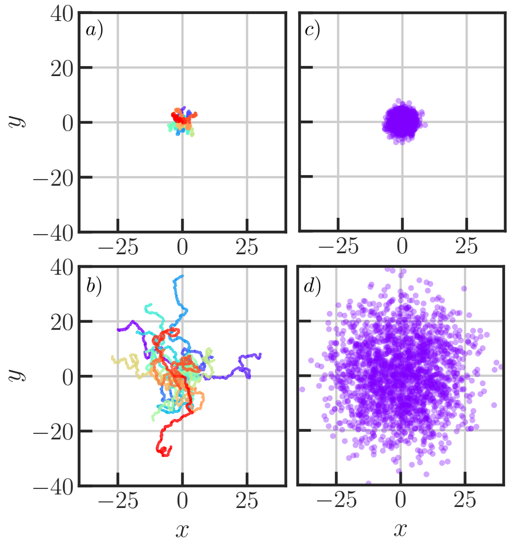

Figure 1.3: Comparison of passive and active Brownian colloid trajectories. Panels **a)** and **c)**: Twenty stacked trajectories obtained from discretising Eqs. 1.3 and 1.4 respectively, with $\Delta t = 0.05$, $n_{\text{steps}} = 5000$, $D = 0.05$, $D_{\text{R}} = 0.3$, $v = 0.1$, with all units in simulation units. Colours are only used for readability and have no physical meaning. Panels **c)** and **d)**: final position of 2000 simulations for the passive and active Brownian particle models, respectively.

speed ω in the angular Langevin equation, as follows [20, 21, 22]:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \omega\sqrt{2D_{\text{R}}}\xi_\phi(t)\ .\end{aligned} \tag{1.6}$$

It is therefore possible to play with the angular velocity ω to obtain different behaviours, from shrinking spirals to circles to expanding spirals for the two-dimensional case.

Three-dimensional, helical motion can be described starting from a three-dimensional ABP model, such as

$$\begin{aligned}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} &= v\mathbf{n} + \sqrt{2D_t}\,\boldsymbol{\xi}_{\mathbf{T}}(t), \\ \frac{\mathrm{d}\mathbf{n}}{\mathrm{d}t} &= \boldsymbol{\Omega}\times\mathbf{n} + \sqrt{2D_r}\,\boldsymbol{\xi}_{\mathbf{R}}(t)\times\mathbf{n},\end{aligned} \tag{1.7}$$

with $\mathbf{r}$ the position vector $\mathbf{r} = (x, y, z)$, and $\mathbf{n}$ the orientation vector, which must be normalised, as $\mathbf{n} = (n_x, n_y, n_z)$, $|\mathbf{n}| = 1$, and $\boldsymbol{\Omega}$ is the angular velocity vector defined as

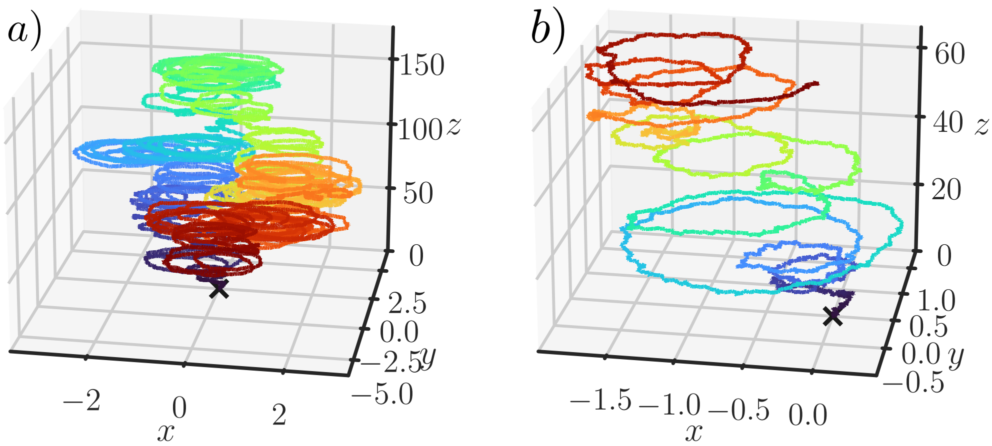

Figure 1.4: Simulated trajectories from the discretised Eqs. 1.7. **a)** Full trajectory for $n_{\mathrm{steps}} = 80000$, $\Delta T = 0.01$, $D = 0.005$, $D_{\mathrm{R}} = 0.01$, $v = 1.0$, $\mathbf{\Omega} = 1.0$ and $\omega = \frac{\pi}{12}$. All units are in simulation units. **b)** Same trajectory, for the first 800 steps. Colourmap corresponds to elapsed time and is only indicative.

$$\mathbf{\Omega} = \omega\, \mathbf{e}_z = \begin{pmatrix} 0 \\ 0 \\ \omega \end{pmatrix} , \tag{1.8}$$

for a rotation about the z axis (Fig. 1.4).

1.2.2 Run-and-tumble

A RTP is an active particle that alternates between two types of motion: long periods of approximately straight swimming, called runs, and sudden reorientation events, called tumbles [23, 24, 25]. During a run, the particle moves at a nearly constant self-propulsion speed v along its instantaneous orientation $\phi(t)$. The position therefore evolves as

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D}\xi_y(t) \ ,\end{aligned} \tag{1.9}$$

which are the same equations as the ABP case. However, the orientation of the particle may suddenly change, usually according to a Poisson process with rate λ. This change is called a tumble, and happens for instance when a *Chlamydomona* or an *E. coli* desynchronises its flagella. The probability that at least one tumble occurs during a given time Δt is

$$P_{\text{tumble}}(\Delta t) = 1 - \exp(-\lambda \Delta t) \ . \tag{1.10}$$

The typical time between two tumbles is then

$$\tau_{\text{run}} = \frac{1}{\lambda} \ . \tag{1.11}$$

At each tumble time t_i, the orientation is instantly changed as

$$\phi(t_i^+) = \phi(t_i^-) + \Delta\phi_i \ , \tag{1.12}$$

where $\phi(t_i^+)$ is the orientation immediately after the tumble, $\phi(t_i^+)$ the orientation immediately before and $\Delta\phi_i$ is the reorientation.

The particle has a persistent direction of motion, but this persistence is destroyed by two mechanisms: continuous rotational diffusion, controlled by D_{R}, and tumbling events, controlled by λ. In two dimensions, for fully random tumbles, the persistence timescale is given by

$$\tau_{\mathrm{p}} = \frac{1}{D_{\mathrm{R}} + \lambda} \ . \tag{1.13}$$

At very long time, the process thus becomes diffusive, much like the ABP case again, and with diffusivity

$$D_{\text{eff}} = D + \frac{v^2}{2(D_{\mathrm{R}} + \lambda)} \ . \tag{1.14}$$

The MSD in that case is given by (see Fig. 1.5)

$$\boxed{\mathrm{MSD}_x(t) = \mathrm{MSD}_y(t) = 2Dt + \frac{v^2}{\lambda^2}\left(\lambda t + e^{-\lambda t} - 1\right)} \tag{1.15}$$

This equation is strictly equivalent to Eq. 1.5, save for a substitution of λ with D_{R}. The

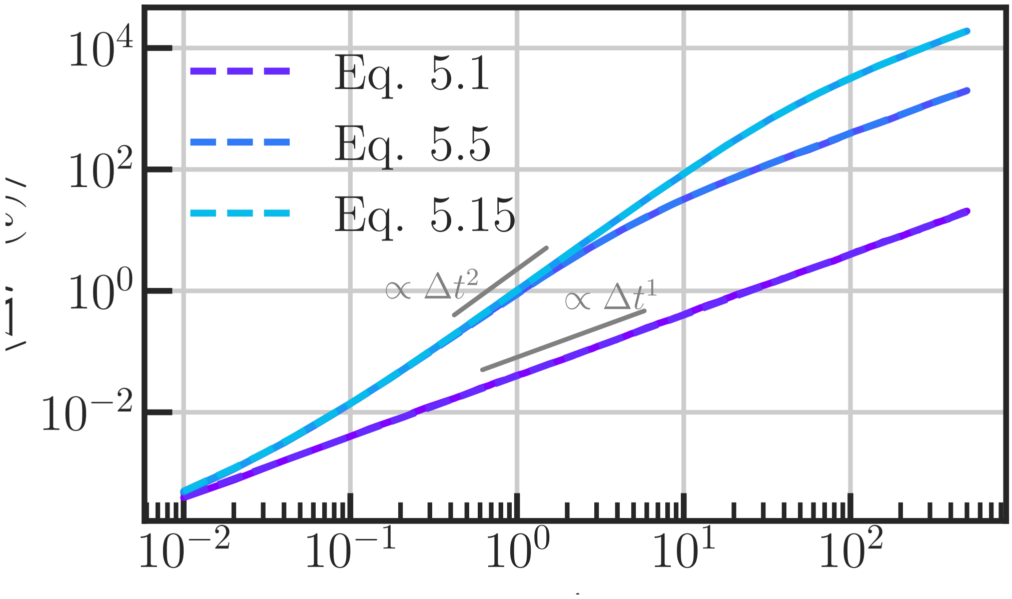

Figure 1.5: MSDs of an ABP (light blue line), a run-and-tumble particle (navy blue line) and a passive Brownian colloid (purple line), for trajectories of $N_{\text{steps}} = 5 \times 10^5$ steps, $\Delta t = 0.01$, $D = 0.01$, $v = 1.0$, and $\lambda = 0.05$. Corresponding theoretical curves from Eqs. 1.3, 1.5 and 1.15 are shown in dashed lines in respective colours.

MSD is not able to distinguish between processes, and one must turn to other quantities (non-gaussianity, velocity autocorrelation, or even visually by trajectory shape) to properly assess the difference between ABP and RTP processes.

Now, we can identify two limiting cases for the RTP process: when $\lambda = 0$, we naturally retrieve the ABP process. When $D_\text{R} = 0$, the motion is a pure run-and-tumble for reorientation comes solely from the active process, with $D_{\text{eff}} = \frac{v_0^2}{2\lambda}$. This is the limit in which we work for the project I present at the end of this Chapter.

1.2.3 Other conventional methods for swimmer modelling

The models I presented so far describe active matter through an ensemble of reduced sets of degrees of freedom. Therein, the swimmer is treated as a point-like particle through its leading far-field hydrodynamic signature. Though minimal, these methods are efficient in terms of computation cost. In the next section, I present a case study where complex swimmers interact with a flow and two boundaries using these techniques and recover the correct behaviour observed in experiments. However, these more elementary techniques do not fully resolve the finite swimmer size, surface motion, full lubrication forces, or detailed flow structures that become important when swimmers come close to one another or to a boundary.

A first step is to represent the swimmer as a finite-size body with surface activity. The classical example is the squirmer model introduced by James Lighthill in 1952 [26] but refined by Blake in 1971 [27], in which a spherical particle swims by imposing a tangential slip velocity on its surface. The propulsion mechanisms are then modeled through the flow generated at the swimmer's boundaries. It is then possible to recover different swimmer behaviours (Shaker, pusher, puller, etc.) by tweaking the system's parameters. This model can be implemented in numerical methods such as the RigidMultiblobWalls method I describe in Chapter 2 to model complex, active bodies in a fluid.

More resolved approaches include solving the fluid, such as boundary-element, immersed-boundary, multiparticle collision dynamics, or lattice-Boltzmann methods. These methods compute the fluid flow around the swimmer and can incorporate complex swimmer shapes, deformable boundaries, confinement, or no-slip walls. They are much more computationally demanding, but they provide access to regimes where the point-particle approximation breaks down.

Point-particle models are well suited to identify generic non-equilibrium mechanisms, and finite-size and hydrodynamically resolved models are necessary when near-field interactions, confinement, or details in the propulsion mechanism are in control of the dynamics.

1.3 Active Brownian Particles in fluid flows

1.3.1 Context

In Chapter 4, I introduce the notion of Taylor-Aris dispersion, originally developed for passive solutions in Poiseuille flows. Extensions of this framework have been developed for biased and gyrotactic swimming microorganisms [28, 29, 30, 31], as well as for idealized active particles and active Brownian particles in confined flows [32, 33]. Our study provides an experimental realization with living *Chlamydomonas reinhardtii* in a confined, sinusoidal Poiseuille flow, which is validated numerically with Langevin simulations. Our study was published in the Proceedings of the National Academy of Sciences of the United States of America (PNAS) in 2025 [34].

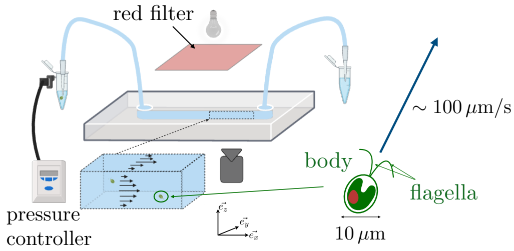

Figure 1.6: Schematic representation of the experimental setup for Active Taylor-Aris dispersion. A suspension of microswimmers is injected in a rectangular microchannel of width W, height H and length L. An oscillating flow is imposed in the microchannel using a pressure controller at one end, whereas pressure is maintained constant on the other end with a reservoir. Swimmers are illuminated with a red-filtered light, and imaged with a high-framerate camera through an objective.

1.3.2 Experimental setup

We create a microfluidic channel using soft lithography on a PolyDiMethylSiloxane (PDMS) chip. The channel has dimensions $(L, W, H) = (88\text{mm}, 180\mu\text{m}, 100\mu\text{m})$, with W the width, H the height and L the length of the channel (Fig. 1.6). The streamwise direction is defined along $\mathbf{e_x}$. The cell is then slowly filled with a dilute suspension of *Chlamydomonas reinhardtii* to ensure negligible interaction between individual algae. We suppress phototacic response of the algae by filtering out light wavelength save for red.

A pressure controller is plugged at one end of the channel, on the entry point. On the exit point, constant pressure is maintained by a hydrostatic bath. We input a controlled pressure, which can be modulated by a sinusoid using the pressure controller. The pressure difference between the entry and exit points launches the flowing of the suspension inside the microchannel. The fluid velocity is given by:

$$\mathbf{u}(y, z, t) = \frac{\Delta P}{R_\mathrm{h}} f(y, z) \sin\left(\frac{2\pi t}{T} + \phi_0\right) \mathbf{e_x} \,, \tag{1.16}$$

where ΔP is the pressure difference between the amplitude of the pressure controller and the hydrostatic bath, R_h is the hydraulic resistance of the channel, T is the pressure oscillation period, and ϕ_0 is a phase at origin, and $f(x, y)$ is the cross-sectional profile, which in a quasi two-dimensional limit is expressed $f(y) = 4\left(\frac{y}{W}\right)\left(1 - \frac{y}{W}\right)$. Because the

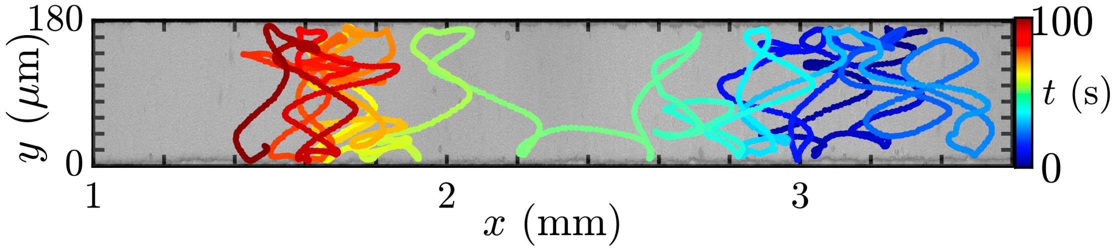

Figure 1.7: Typical experimental trajectory of a *Chlamydomonas reinhardtii* in an oscillating flow. The colormap corresponds to elapsed time.

channel is sufficiently narrow and our observations flatten out the z coordinate, we use this expression of f as our approximation for the study.

We then image the channel from the top, over the $z-$plane. We record frames for 200 s at 40 frames per second. We then link individual trajectories using a linking algorithm, and only keep trajectories which last for more than 20 s in the field of view of the camera for treatment. The oscillatory flow has the advantage of forcing the algae into the field of view for longer durations (Fig. 1.7).

The treatment of a single trajectory then involves demodulating the signal from the oscillatory signature of the flow velocity using a sliding windowed demodulation, assuming that the position of an alga in time along the $x-$axis can be expressed as

$$x(t) = \hat{x}(t) + A[y(t), t] \cos\left(\frac{2\pi t}{T}\right) \ , \tag{1.17}$$

where $A[y(t), t]$ is an unknown function, and $\hat{x}(t)$ is the position of the alga in the flow's comoving frame.

1.3.3 Numerical model

To reproduce the experimental results from this study, I choose a two-dimensional RTP model with a reorientation rate $\tau_{\mathrm{R}}^{-1} = D_{\mathrm{R}}$. Strictly speaking, this D_{R} is not a diffusion coefficient in the thermal sense, but for the sake of continuity with experimental nomenclature, we still define it as an active rotational diffusion coefficient. The equations thus give:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v\cos[\phi(t)] + \sqrt{2D}\xi_x + u(y,t)\ , \tag{1.18}$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = v\sin[\phi(t)] + \sqrt{2D}\xi_y\ , \tag{1.19}$$

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{1}{2}\omega(t) + \sqrt{2D_\phi}\xi_\phi\ , \tag{1.20}$$

with v the particle's self propelling speed, $D_\phi = \tau_{\mathrm{R}}^{-1}$ the rotation rate, ω the rotational velocity due to the vorticity of the flow, and the fluid velocity is expressed, strictly equivalently to Eq. 1.16, as $\mathbf{u}(y,t) = \frac{1}{2\eta_0}\frac{\Delta P}{L}y(H-y)\sin\left(\frac{2\pi t}{T}\right)\mathbf{e_x}$, with η_0 the fluid viscosity.

I then disretise Eqs. 1.20 using a forward Euler scheme. The results of solving the discretised Eqs. 1.20 is shown in Figs. 1.8 and 1.9. By comparing a case where particles are active but in no advection, and a case where they move both due to their own activity and to the fluid flow, one notices a very noticeable enhancement in the dispersion (Fig. 1.8). Indeed, in absence of flow, particles tend to quickly reach one boundary, and then take a noticeable amount of time (of the order of τ_{R}) to reorient away from the wall through their active rotational process. In the case where a Poiseuille flow exists, particles tend not to approach the wall as much. I interpret this as a direct result of the stronger shear close to the wall: a swimmer will only need a tiny"pus" in its orientation to escape contact with the wall.

Along the $x-$axis, we see that the trajectory of a swimmer presents a strong signature of the oscillations of the fluid flow (Fig. 1.9). To demodulate the signal, I use a different technique than the experimentalists on this study, with a lower computational cost: a simple linear regression. I know that the parasitic drift I have to remove has a known frequency, which I choose when running the simulations. I feed to a simple linear regression model all the parameters I know, as well as the smoothed $y-$ coordinate for each time step to avoid overfitting thermal noise. This is important, as we see in Eq. 1.16 that the amplitude of the sine wave depends on the $y-$position of the particle. Finally, the model takes the $x-$positions and returns a demodulated version, which we call $\hat{x}(t)$.

Along the $y-$axis, I simply make sure before conducting momentum analyses that the particle has had enough time to explore the whole channel cross-section fully, multiple times. This is vital to ensure the particle has crossed the Poiseuille streamlines, a condition which ensures we are in the Taylor-Aris regime for passive Brownian colloids.

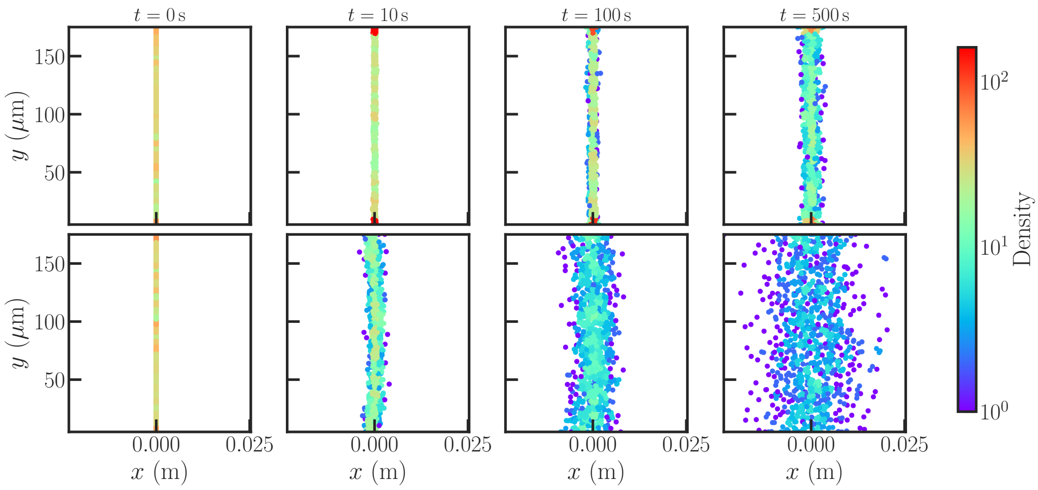

Figure 1.8: Snapshots of 100 numerically generated trajectories obtained from discretising Eqs. 1.20 with an Euler scheme with $N_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\mathrm{s}$, $T = 2/,\mathrm{s}$, $k_{\mathrm{B}}\Theta = 4.14 \times 10^{-19}\,\mathrm{J}$, $v = 100 \times \mathrm{\mu m.s^{-1}\,m.s^{-1}}$, $D_{\mathrm{R}} = 3.35\,\mathrm{s}^{-1}$, $a = 5\,\mathrm{\mu m}$ for the active case without a flow (top), where $\Delta P = 0.0\,\mathrm{Pa}$, and with a flow (bottom), where $\Delta P = 73.0\,\mathrm{Pa}$. The simulations do not account for interactions, we assume here a perfect gas behaviour for the swimmers. The colormap corresponds to the local density of ABPs in a binned (x, y) space. The Taylor-Aris dispersion effect is clearly visible by comparing the top and bottom plots at each given time, as we see the bulk of ABPs spreading for the active, advected case, much more than for the purely active case.

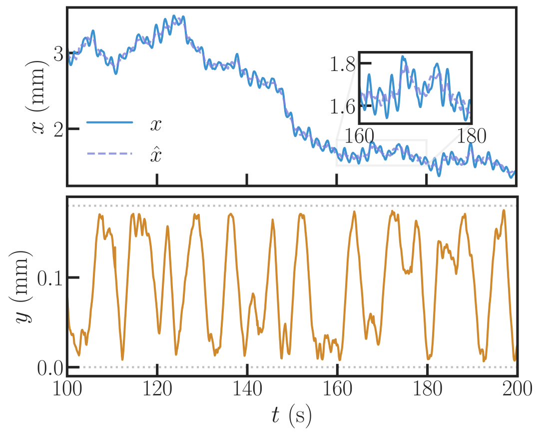

Figure 1.9: Numerical results obtained from discretising Eqs. 1.20 with an Euler scheme with $N_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\text{s}$, $T = 2/,\text{s}$, $k_{\text{B}}\Theta = 4.14 \times 10^{-19}\,\text{J}$, $v = 100 \times \,\mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_{\text{R}} = 3.35\,\text{s}^{-1}$, $\Delta P = 73.0\,\text{Pa}$, $a = 5\,\mu\text{m}$. Top: streamwise trajectory along the $x-$axis. The full blue curve shows the raw trajectory, with a strong modulation from the fluid flow oscillations. The indigo dashed curve is the result of a demodulation of the signal obtained from a linear regression. The inset shows a zoom on times $t \in [160; 180]$ for better visualiation of the demodulation. Bottom: cross-streamwise trajectory along the $y-$axis (orange curve). The grey dotted lines represents the boundary of the channel, beyond which the ABP cannot wander. Multiple explorations of the full available width are visible, which is a key observation to ensure the ABP has crossed the full flow profile multiple times and can therefore be considered in a Taylor-Aris dispersion regime.

1.3.4 Results

We recall the Taylor-Aris dispersion formula from the last Chapter, but keep in mind that the swimmer now has an effective diffusion coefficient at long times which is not D, the thermal diffusion coefficient, but D_0, which we defined in the above sections as $D_0 = v^2/D_\mathrm{R}$. The expected enhancement following the Taylor-Aris prediction thus reads

$$D_{x,\mathrm{TA}} = D_0(1 + \alpha \mathrm{Pe}^2) \ , \tag{1.21}$$

where $D_{x,\mathrm{TA}}$ is the expected dispersion coefficient for a Brownian colloid with diffusion coefficient D_0, α is a geometrical factor depending on the channel geometry (here, $\alpha = \frac{1}{210}$) and Pe is an active Péclet number which compares advection and diffusion, as

$$\mathrm{Pe} = \frac{\Delta P}{\sqrt{2} R_\mathrm{h}} \frac{W}{D_0} \ , \tag{1.22}$$

where the $\frac{\Delta P}{\sqrt{2}R_\mathrm{h}}$ factor is the root mean square fluid velocity due to the oscillations of the pressure difference.

To extract diffusion coefficients and dispersion along the $x-$axis, we compute the dedridfted MSDs of multiple trajectories for a given set of parameters and average them. This is not the most computationally efficient method, but it is sufficient as we do not require a large amount of statistics for this study. We then repeat this process for multiple ΔP values, or equivalently, multiple Pe values. The results of these calculations is shown in Fig. 1.10.

We already notice that every MSD follows the expected shape from Eq. 1.15, in a case where the thermal diffusion coefficient D is too small to be visible at very short times. The first visible regime is therefore the ballistic regime, where the particle hasn't had time yet to reorient. Beyond the typical reorientation time τ_R, the MSDs all take a diffusive behaviour. Furthermore, as the Péclet number grows, MSDs retain their shape but are systematically higher, which is the hallmark of an enhanced dispersion coefficient due to fluid flow advection.

We now turn to quantifying that enhancement of the dispersion. I present in Fig. 1.11 a typical plot in which dispersion coefficients are measured, from performing log-fits in the diffusive regime of the $\hat{x}-$wise MSD. In this treatment, I also retrieve the ballistic prefactors at short times, which correspond to the mean squared velocity of the particle in the $x-$ and $y-$ directions for later analyses.

Finally, the treatment is repeated over a span of Péclet numbers. The enhancement on the diffusivity D_x/D_0 is plotted against Pe^2 in Fig. 1.12. We notice that the dispersion

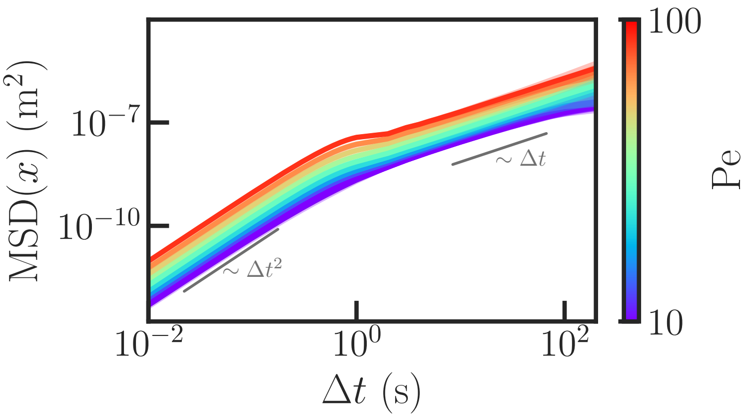

Figure 1.10: MSDs along the $x-$axis computed from demodulated numerical trajectories for various Péclet numbers, with $N_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\text{s}$, $T = 2/,\text{s}$, $k_\text{B}\Theta = 4.14 \times 10^{-19}\,\text{J}$, $v = 100 \times \,\mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_\text{R} = 3.35\,\text{s}^{-1}$, $a = 5\,\mu\text{m}$. The colormap corresponds to the Péclet number of a given numerical experiment. The grey lines indicate a slope of 2 on the left and 1 on the right in log-scale, which correspond to the expected ballistic regime at short times and effective diffusive regime at long times of an ABP. Higher Péclet numbers show both a stonger ballistic regime and a larger diffusivity at long times.

enhancement follows the trend set by Eq. 1.21, but may become smaller or larger than a passive case by tuning the fluid oscillation period to lower or higher periods respectively. This hints at an asymptotical behaviour for very large periods, where the enhancement of the diffusivity reaches a higher profile which still follows the trend set by Eq. 1.21 as $D_x/D_0 = 1+\alpha\text{Pe}^2$. However, the α coefficient is modified by the activity of the particle, which explains why the active Taylor dispersion phenomenon reaches higher values than its passive counterpart.

The mean squared velocities (Fig. 1.13) on the other hand do not demonstrate a strong effect of the fluid oscillation period. This is expected as we chose our oscillation period much greater than τ_R, which means that in the ballistic regime, the particle never has time to feel the fluid oscillation before having reoriented. We also notice that $\langle V_y^2\rangle$ remains unchanged with Péclet number, whereas $\langle V_x^2\rangle$ increases with Pe^2. This is consistent with speed composition, and we expect the ratio $\langle V_x^2\rangle/\langle V_y^2\rangle$ to follow

$$\langle V_x^2\rangle/\langle V_y^2\rangle = 1 + \beta\text{Pe}^2\ , \tag{1.23}$$

where β is a geometric factor. Couldnt PeA just be included in beta ? what is PeA anyway ? Pe already is an active Péclet. I ask Marc now.

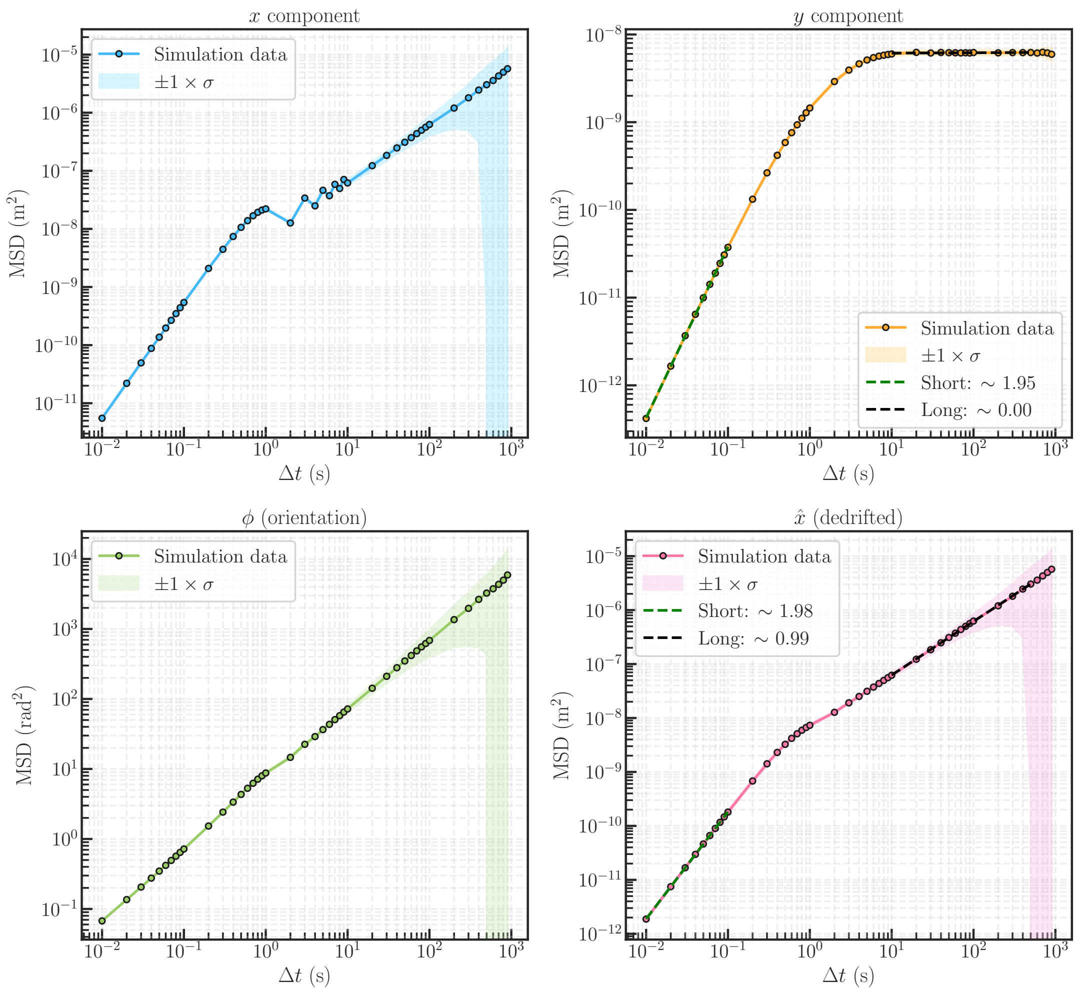

Figure 1.11: Typical MSDs computed from 100 stacked trajectories and averaged, with $N_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\text{s}$, $T = 2/,\text{s}$, $k_B\Theta = 4.14 \times 10^{-19}\,\text{J}$, $v = 100 \times \mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_R = 3.35\,\text{s}^{-1}$, $a = 5\,\mu\text{m}$, $\Delta P = 10.0\,\text{Pa}$. The blue curve corresponds to the MSD along the $x-$axis, and presents a storng signature of the flow's oscillation. The orange curve corresponds to the $y-$axis. It presents two regimes: a ballistic regime at short times (with an exponent of 2), and a confined regime at long times (with an exponent of 1). The green curve corresponds to the angular MSD, which remains diffusive at all times, as we can expect from the angular Eq. 1.20. Finally, the pink curve is the demodulated MSD on the $x-$axis. At short times, it also presents a ballistic regime, and crosses over to a diffusive regime at long times (with an exponent of 1). All fits are performed in log-scale and all D_x, $\langle V_x^2 \rangle$ and $\langle V_y^2 \rangle$ values are extracted from dataset such as the one shown here.

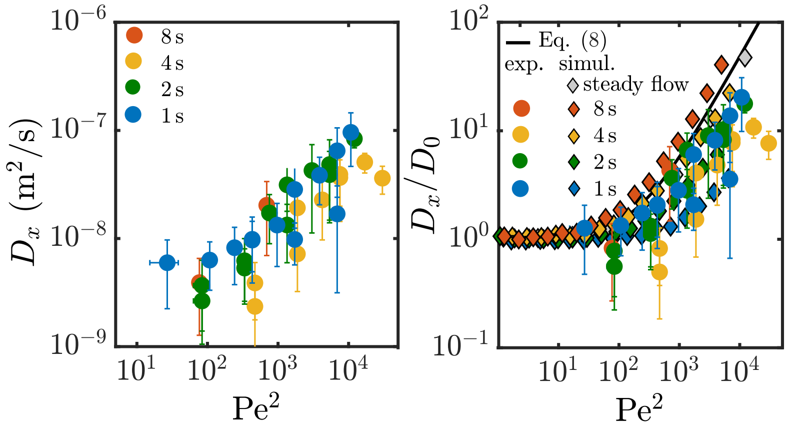

Figure 1.12: Measured enhacement of the streamwise diffusion D_x/D_0 of an ABP in an oscillating poiseuille flow as a function of Péclet number. **a)** Experimental dispersion coefficients D_x. **b)** Comparison of experimental and numerical renormalised dispersion coefficients D_x/D_0 against Pe^2. Grey markers correspond to a passive Brownian tracer with the same parameters otherwise. Diamonds are the numerical simulation results, whereas circles correspond to the experimental data. The black curve shows the expected enhancement of the dispersion from Eq. 1.21. Colours correspond to various fluid oscillation periods.

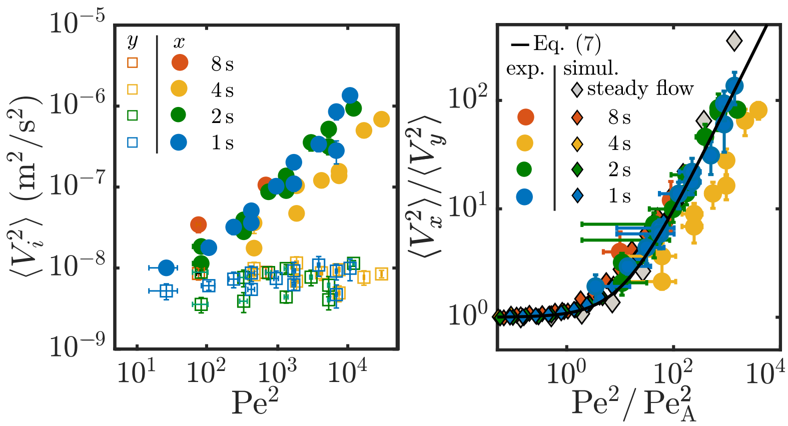

Figure 1.13: Measured enhacement of the streamwise mean squared velocities $\langle V_x^2 \rangle / \langle V_y^2 \rangle$ of an ABP in an oscillating poiseuille flow as a function of Péclet number. **a)** Experimental measurements of the mean squared velocities along $\mathbf{e_x}$ and $\mathbf{e_y}$. **b)** comparison of experimental and numerical measurements of the ratio $\langle V_x^2 \rangle / \langle V_y^2 \rangle$ against Péclet number squared. Grey markers correspond to a passive Brownian tracer with the same parameters otherwise. Diamonds are the numerical simulation results, whereas circles correspond to the experimental data. Colours correspond to various fluid oscillation periods. Black line is Eq. 1.23.

1.4 Conclusion and perspectives

In this Chapter, I have investigated the dispersion of active microswimmers in confined viscous flows, using *Chlamydomonas reinhardtii* as a model swimmer. The aim was to test whether a classical theory for passive transport, namely the Taylor-Aris dispersion, can be extended to self-propelled particles whose motion is persistent and orientation variable. For this, microfluidic experiments were combined with single-cell tracking, statistical analysis, numerical Langevin simulations, and theoretical modelling.

The experiments confirmed that the spatial distribution of the alga is strongly affected by confinement. In agreement with previous observations on swimming microorganisms, the cells were found to accumulate near the channel walls. This boundary accumulation is not a geometric detail: it determines how the swimmers sample the imposed Poiseuille flow and it plays a central role in their cross-streamwise distribution. In the direction along the flow, the MSD displayed a clear crossover from a short-time ballistic regime, dominated by persistent swimming and flow advection effects, to a long-time diffusive-like regime. The latter regime was clearly affected by the amplitude of the imposed flow.

These observations were quantified by a minimal theoretical description and by RTP Langevin simulations. Despite the complexity of the living system, the main behaviours could be captured by an effective active-particle description.

Several directions remain open. First, *Chlamydomonas reinhardtii* is a puller-type swimmer, and it would be interesting to test whether the same active Taylor-Aris mechanism persists, or is modified and how, for pusher-type microorganisms such as *E. coli*. Second, the present work focused on unbiased swimming in imposed flows. Incorporating external stimuli such as light, chemical gradients, or electromagnetic fields in general could reveal how taxis modifies active dispersion. Stronger confinement would require a more detailed treatment of near-wall hydrodynamic and lubrication effects, as well as possible transitions between swimming, adhesion, and gliding. Finally, the oscillatory flows used here were kept in a quasi-steady regime to obtain sufficient statistical resolution. Exploring higher oscillation frequencies, especially beyond the natural orientational decorrelation rate τ_R of the swimmers, could reveal new couplings which are relevant to biological settings.

Last thing to do: I asked Marc what PeA is on his figures, because it is unclear to me why we need it (3.7 factor ??). Will update when i have understood. Sorry tom. Code is ready. Add chlamy github to chapter. Add code to git thesis.

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